WEEK 12

AUTOMOBILE ANALYTICS DASHBOARD

Introduction :

Automobiles have different features such as price, horsepower, engine size, fuel efficiency, and body style. Comparing these features helps to understand the differences between various automobile manufacturers.

In this project, an Automobile Analytics Dashboard was developed using Microsoft Power BI. The preprocessed Automobile Dataset from Week 11 was used to create interactive charts, KPI cards, comparison matrices, and custom tooltips.

The dashboard helps to analyze automobile prices, engine performance, fuel efficiency, and technical specifications in a simple visual format.

Objectives :

1. To compare automobile manufacturers based on price and performance.
2. To analyze the relationship between horsepower, engine size, and price.
3. To compare automobile prices across different body styles.
4. To analyze city and highway fuel efficiency.
5. To classify automobiles into Budget, Mid-Range, and Luxury segments.
6. To compare body styles and drive-wheel types using a matrix.
7. To create custom tooltips to display additional vehicle details.

Dataset Description :

The dataset contains automobile information such as:

1. Make
2. Price
3. Horsepower
4. Engine Size

5. City MPG
6. Highway MPG
7. Body Style
8. Drive Wheels
9. Fuel Type
10. Fuel System
11. Curb Weight
12. Wheel Base
13. Number of Cylinders

The data cleaning and preprocessing activities were completed in Week 11. The prepared dataset was used directly for Week 12 dashboard development.

Dashboard Development :

The dashboard was developed using Microsoft Power BI. Three report pages were created to present automobile information and comparisons.

MAIN AUTOMOBILE DASHBOARD :

KPI Cards :

Four KPI Cards were created to display important automobile statistics.

1. Average Price

- Field: price_filled
- Calculation: Average
- Purpose: Displays the average price of automobiles.

2. Average Horsepower

- Field: horsepower_filled
- Calculation: Average
- Purpose: Displays the average horsepower of automobiles.

3. Average Engine Size

- Field: engine-size
- Calculation: Average
- Purpose: Displays the average engine size.

4. Average Highway MPG

- Field: highway-mpg
- Calculation: Average
- Purpose: Displays the average highway fuel efficiency.



Scatter Chart – Horsepower vs Price :

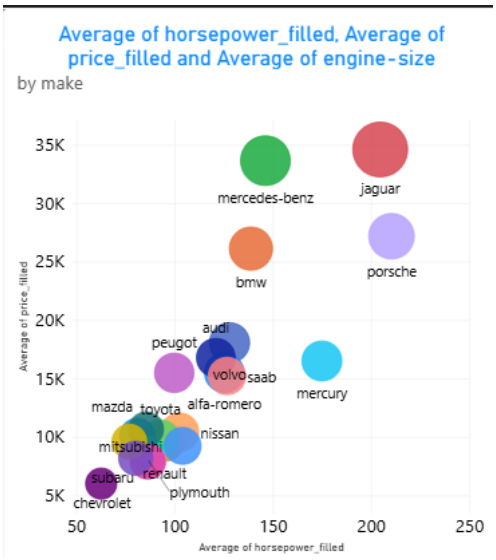
Visual Used: Scatter Chart

Parameters:

1. X-Axis: horsepower_filled – Average
2. Y-Axis: price_filled – Average
3. Size: engine-size – Average
4. Values: make

The scatter chart was created to compare automobile manufacturers based on horsepower and price. The bubble size represents the average engine size. Each manufacturer is represented in the chart.

This chart helps to understand the relationship between automobile performance and price.



Column Chart – Average Price by Body Style :

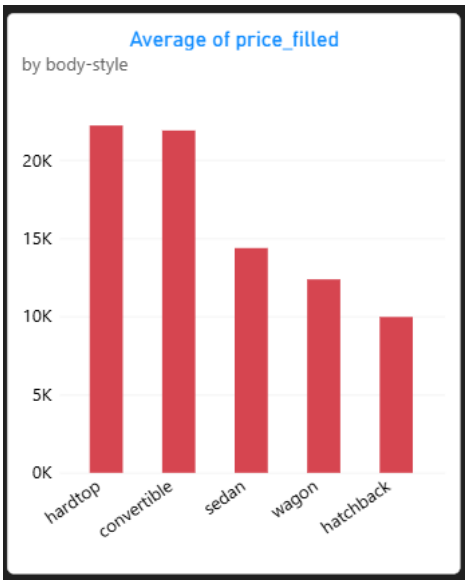
Visual Used: Clustered Column Chart

Parameters:

1. X-Axis: body-style
2. Y-Axis: price_filled – Average

This chart compares the average price of different automobile body styles, such as Sedan, Hatchback, Wagon, Convertible, and Hardtop.

It helps to identify the price differences between various body styles.



AUTOMOBILE COMPARISON DASHBOARD :

Matrix – Body Style vs Drive Wheels :

Visual Used: Matrix

Parameters:

1. Rows: body-style
2. Columns: drive-wheels
3. Values: price_filled – Average

The matrix was created to compare automobile prices based on body styles and drive-wheel types.

It displays different body styles in rows and drive-wheel types in columns. The average price is displayed for each combination.

This helps to understand price differences between various body styles and drive-wheel categories.

body-style	4wd	fwd	rwd	Total
convertible		11595.00	23949.60	21890.50
hardtop		8249.00	24202.71	22208.50
hatchback	8949.00	8396.39	14125.00	9967.09
sedan	12647.33	9828.75	21711.83	14372.99
wagon	9095.75	9997.33	16994.22	12371.96
Total	10247.00	9262.28	19633.11	13150.31

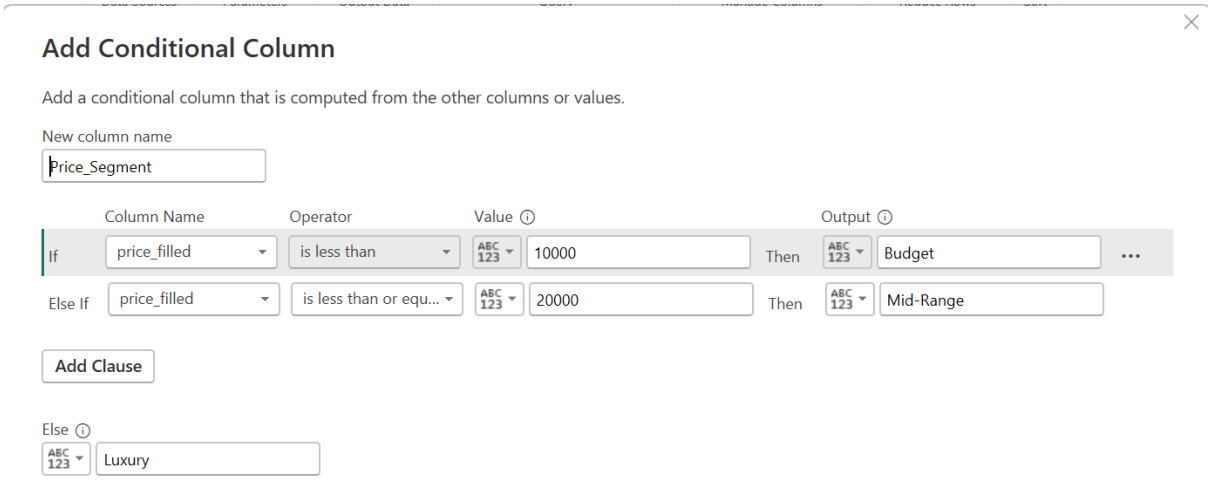
Conditional Column – Price Segment :

A conditional column named Price Segment was created using the price_filled field.

The automobiles were divided into three categories:

1. Budget – Price below 10,000.
2. Mid-Range – Price from 10,000 to 20,000.
3. Luxury – Price above 20,000.

This column was created to classify automobiles based on their price.



Manufacturer vs Price Segment Chart :

Visual Used: Stacked Column Chart

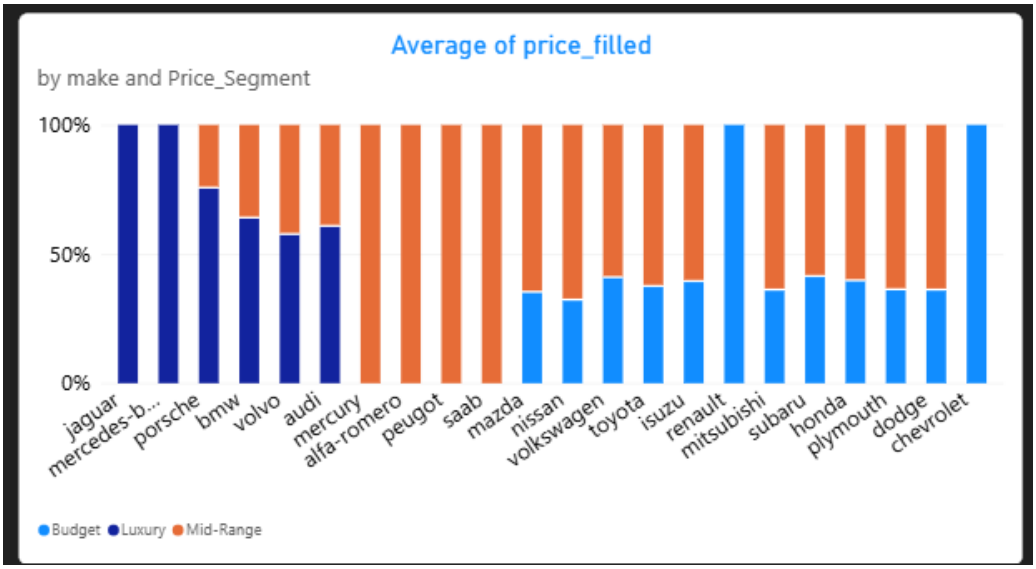
Parameters:

1. X-Axis: make
2. Y-Axis: price_filled – Average
3. Legend: Price Segment

This chart was created to compare different automobile manufacturers across Budget, Mid-Range, and Luxury price categories.

The Price Segment conditional column was used as the legend to display the different price categories.

It helps to understand the price categories of different automobile manufacturers.



City MPG vs Highway MPG :

Visual Used: Clustered Column Chart

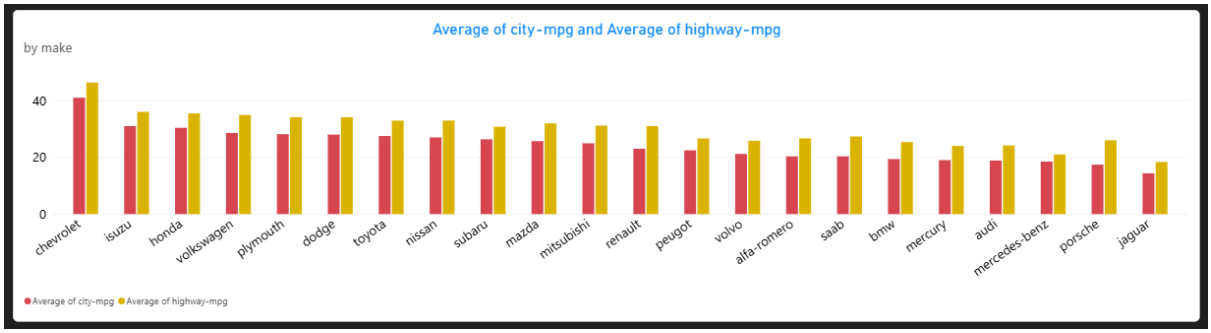
Parameters:

1. X-Axis: make
2. Y-Axis: city-mpg – Average
3. Y-Axis: highway-mpg – Average

This chart compares the average city fuel efficiency and average highway fuel efficiency of different automobile manufacturers.

City MPG represents fuel efficiency during city driving, while Highway MPG represents fuel efficiency during highway driving.

It helps to identify the differences in fuel efficiency between city and highway driving.



CUSTOM TOOLTIP :

Tooltip Page Creation :

A separate report page was created for displaying additional automobile details. The page was configured as a tooltip page by enabling the Allow use as tooltip option.

Five KPI Cards and one technical details table were added to this page.

KPI Cards in Tooltip :

The following five KPI Cards were created:

1. Average Price

- Field: price_filled
- Calculation: Average

2. Average Highway MPG

- Field: highway-mpg
- Calculation: Average

3. Average Engine Size

- Field: engine-size
- Calculation: Average

4. Average City MPG

- Field: city-mpg
- Calculation: Average

5. Average Horsepower

- Field: horsepower_filled
- Calculation: Average

These cards display additional automobile information in the custom tooltip.

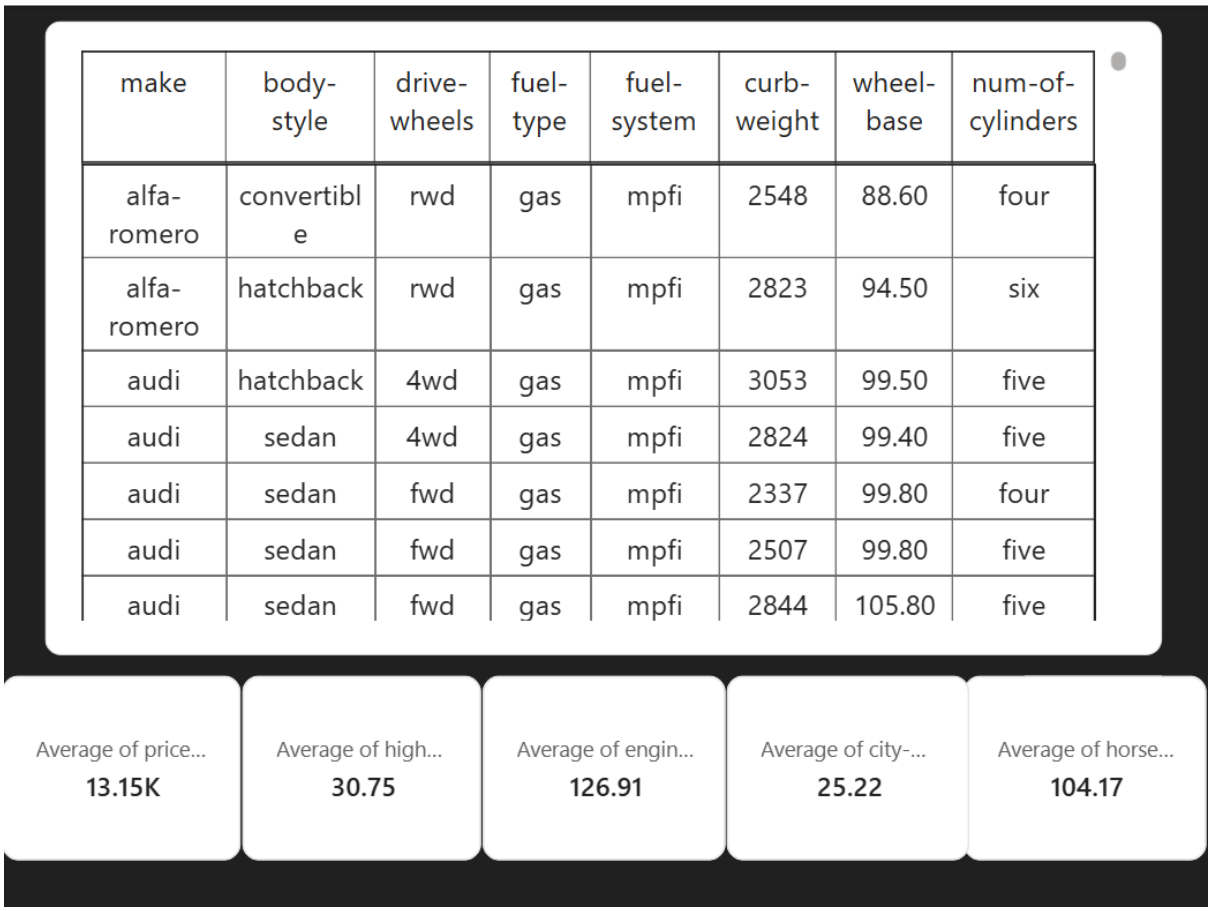
Vehicle Technical Details Table :

A Table visual was created to display the technical specifications of automobiles.

The following fields were added:

1. make
2. body-style
3. drive-wheels
4. fuel-type
5. fuel-system
6. curb-weight
7. wheel-base
8. num-of-cylinders

The table provides additional information about the automobile's manufacturer, body style, fuel system, weight, wheel base, and engine specifications.



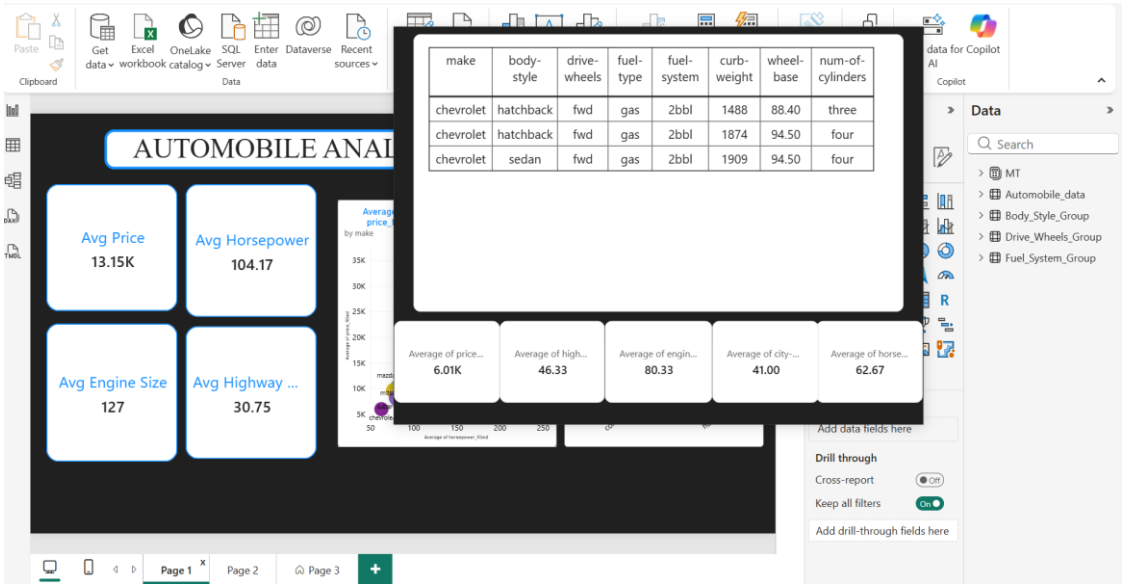
make	body-style	drive-wheels	fuel-type	fuel-system	curb-weight	wheel-base	num-of-cylinders
alfa-romero	convertible	rwd	gas	mpfi	2548	88.60	four
alfa-romero	hatchback	rwd	gas	mpfi	2823	94.50	six
audi	hatchback	4wd	gas	mpfi	3053	99.50	five
audi	sedan	4wd	gas	mpfi	2824	99.40	five
audi	sedan	fwd	gas	mpfi	2337	99.80	four
audi	sedan	fwd	gas	mpfi	2507	99.80	five
audi	sedan	fwd	gas	mpfi	2844	105.80	five

Applying and Testing the Custom Tooltip :

The custom tooltip page was connected to the Scatter Chart on Page 1.

When the mouse pointer is moved over a data point in the Scatter Chart, the tooltip displays additional automobile information.

The tooltip functionality was tested successfully.



make	body-style	drive-wheels	fuel-type	fuel-system	curb-weight	wheel-base	num-of-cylinders
chevrolet	hatchback	fwd	gas	2bbl	1488	88.40	three
chevrolet	hatchback	fwd	gas	2bbl	1874	94.50	four
chevrolet	sedan	fwd	gas	2bbl	1909	94.50	four

Conclusion :

The Automobile Analytics Dashboard was successfully developed using Microsoft Power BI. The dashboard provides a clear comparison of automobile manufacturers based on price, horsepower, engine size, fuel efficiency, body style, and other technical specifications.

Different visualizations such as KPI Cards, Scatter Charts, Column Charts, and Matrix were used to present the information. A custom tooltip was also created and successfully tested to display additional vehicle details.

This project helped to understand automobile data analysis and interactive dashboard development using Power BI.